

Model Comparison Report

Forecasting Botswana Food Price Inflation (FAO series 23014) — IndabaX Botswana 2026 Hackathon, Phase 1

1. Models Compared

The submission compares the two required model families side by side — a classical ARIMAX(0,1,2) and a deep learning LSTM — implemented in separate, clearly labelled notebooks, with a gradient-boosting benchmark kept as supporting out-of-sample evidence. All start from the same cleaned monthly table (*Data_Clean/prediction_data.csv*: 276 rows, 2001-01 to 2023-12, zero missing values), target *FAO_23014* (food inflation, %YoY). The two main models deliberately embody different philosophies: the classical model is a parsimonious, transparent benchmark restricted to external macroeconomic drivers (BDI volatility, Brent, policy rate), while the LSTM additionally exploits the related food-price indices (*FAO_23012*, *FAO_23013*) — which pre-training Random Forest importance analysis identified as the two most influential predictors, together exceeding 82% of total importance — to test whether richer multivariate interactions add forecasting value.

2. Architecture and Hyperparameters

Element	Classical — ARIMAX (submitted forecaster)	Deep learning — LSTM
Architecture	ARIMA(0,1,2) with exogenous regressors: first differencing (d=1) with two moving-average terms; exogenous set = <i>BDI_std</i> lag 7, policy rate lag 12, Brent lag 16, monthly seasonal dummies (one reference month dropped), structural-break dummies (2007, 2009, 2022)	Two stacked LSTM layers of 64 hidden units each, followed by a fully connected output layer producing the next-month forecast
Key hyperparameters	Order (p,d,q) = (0,1,2), selected by AIC grid search over candidate (p,d,q) combinations; exogenous lags fixed by pre-whitened CCF and Granger-causality analysis, not tuned	Input features 6; sequence length 12 months (chosen over 6 and 18 by validation RMSE); hidden units 64; layers 2; dropout 0.20
Uncertainty	95% prediction intervals from the fitted model	Monte Carlo Dropout at inference: 100 stochastic forward passes yielding predictive mean, standard deviation, 95% confidence and prediction intervals
Explainability	Intrinsic — coefficients, chosen lags and dummies map to named economic channels	Post-hoc SHAP summary plot quantifying each driver's contribution

3. Training Procedure

ARIMAX was estimated by maximum likelihood after a full diagnostic workflow: seasonal decomposition, change-point detection (RBF/L2), regression-based stationarity testing with break indicators, correlation and pre-whitened CCF analysis, Granger-causality confirmation of every exogenous lag, and VIF screening (the contemporaneous policy rate and one break dummy were removed; Brent lag-16 retained despite a slightly elevated VIF for its demonstrated forecasting relevance). **The LSTM** was trained with mini-batch gradient-based optimisation with validation monitoring throughout: adaptive learning-rate scheduling (the training log records multiple automatic reductions as validation performance stabilised), automatic checkpointing with training-history logging (the final run resumed from epoch 25's checkpoint), and early stopping that halted training when validation performance stopped improving and restored the best-performing checkpoint for evaluation. Feature and target scalers were fitted on the training partition only and persisted as .joblib artifacts for reproducible inference. **The gradient-boosting benchmark** (*HistGradientBoostingRegressor*) was fitted on 38 engineered lag, seasonal and rolling-mean features.

4. Validation Strategy

All learned-model evaluation is chronological — never random — so errors simulate forecasting genuinely unseen months. The LSTM uses a 70/15/15 chronological split (train 2001–2017, validation 2018–2020, test 2021–2023), with sequences built within each partition after splitting so no window straddles a boundary; the validation partition drives learning-rate scheduling, early stopping and window-length selection, and the test partition is touched only once, after the best checkpoint is restored. The gradient-boosting benchmark holds out the final 24 months (2022-01 to 2023-12). The ARIMAX notebook reports in-sample fit, with its pipeline oriented toward the forward forecast; a **persistence** rule (next month = this month) and a **seasonal naive** rule (next month = same month last year) provide honest yardsticks.

5. Per-Model Results (RMSE and companion metrics)

The deep-learning pipeline's own test evaluation compares the LSTM against persistence on identical test months and scale, making this pair directly comparable:

Model (LSTM pipeline test set, 2021–2023)	MAE	MSE	RMSE	R2
Persistence baseline	0.6671	1.2576	1.1214	0.9570
LSTM (best checkpoint)	1.3744	2.5352	1.5922	–2.2222

In original percentage-point units of the %YoY target, the remaining artifacts give:

Model / baseline	Window	Evaluation type	MAE	RMSE	Other
ARIMAX(0,1,2)	Fitted sample	In-sample fit	0.6484	1.1682	sMAPE 15.88%

Model / baseline	Window	Evaluation type	MAE	RMSE	Other
Gradient Boosting benchmark	2022-01 to 2023-12 (24 mo)	Out-of-sample hold-out	1.6494	2.1584	R2 0.8128; bias 0.2373
Persistence baseline	2021-01 to 2023-12 (36 mo)	Out-of-sample	1.4593	2.1571	bias 0.0690
Seasonal naive baseline	2021-01 to 2023-12 (36 mo)	Out-of-sample	6.3642	7.5079	bias 2.7361
LSTM (inverse-scaled export)	2021-01 to 2023-12 (36 mo)	Out-of-sample test	3.6084	4.5113	bias 2.8524

The ARIMAX row shows goodness of fit, not hold-out accuracy, and is deliberately not ranked against the out-of-sample rows. The out-of-sample rows share units but differ slightly in window (24 vs 36 months); both cover the difficult post-2021 inflation period, and the model ranking is unchanged under either window.

6. Residual Diagnostics

ARIMAX. Residual structure was addressed by design: regression-based stationarity testing (with temporary break indicators) established that differencing was needed only once; the two moving-average terms absorb short-run residual autocorrelation left after differencing; and the structural-break dummies remove the 2008- and 2022-era level shifts so residuals are not dominated by regime changes. In-sample errors average roughly 0.65 percentage points. Not all individual coefficients are significant at the 5% level — noted openly, with the model justified on predictive rather than purely inferential grounds. **LSTM.** The residuals reveal the failure mode precisely: on the inverse-scaled test export the mean error (bias) is +2.85pp against an MAE of 3.61pp — errors are overwhelmingly one-signed, meaning the network systematically under-predicted the post-2021 inflation level rather than scattering around it. This is a level-extrapolation failure consistent with the 2022 structural break identified in the classical workflow: the network never saw comparable inflation levels in its 2001–2017 training window. The negative test R2 (–2.22) confirms residual variance exceeding that of a mean predictor. Monte Carlo Dropout intervals quantify this uncertainty rather than hiding it. **Benchmarks.** Persistence residuals are near-unbiased (+0.07pp), reflecting the strong month-to-month continuity of the target; the gradient-boosting benchmark is also close to unbiased (+0.24pp), indicating its errors are genuine noise around the truth rather than systematic drift.

7. Honest Analysis: Which Model Won, and Why

The classical approach won. The LSTM lost to the persistence baseline on its own test set (RMSE 1.5922 vs 1.1214; negative R2) — reported openly, as the evaluation was designed to allow. The reasons are structural, not incidental. First, **data volume:** 276 monthly observations, of which 193 train the network, is far below what a two-layer, 64-unit recurrent architecture needs to learn temporal structure more robustly than explicit statistical modelling. Second, **the target's character:** Botswana food inflation is highly persistent (persistence alone achieves R2 0.957 on test months), so the bar any learned model must clear is unusually high. Third, **the 2022 regime shift:** the test window sits after a structural break the network could not extrapolate from pre-break training data, producing the systematic under-prediction documented above — whereas the classical workflow handles the break explicitly with dummy variables. ARIMAX is submitted as the final forecaster because every exogenous input earned its place through measured cross-correlation and Granger causality, breaks and seasonality are modelled explicitly, and it carries the only complete forward pipeline with calibrated 95% intervals. Its known limitation — ARIMA-family forecasts converge smoothly toward their long-run mean over long horizons — is acknowledged. The LSTM remains valuable as engineering: a reproducible sequence pipeline with checkpointing, Monte Carlo Dropout uncertainty and SHAP explainability, and its result is itself a substantive finding about matching model complexity to data volume.

8. Forecast Generation for the Submission

No realised post-2023 feature values exist, and none are used. The submitted forecast uses a **two-stage procedure**: stage one fits independent SARIMA models to forecast each exogenous driver (BDI volatility, Brent, policy rate) across the horizon; stage two assembles those forecasts into the future lagged exogenous matrix — with seasonal dummies and break indicators — and supplies it to the fitted ARIMAX model, which outputs the food-inflation forecast with 95% prediction intervals. Because the exogenous lags are long (7/12/16 months), much of the early horizon is driven by *observed* 2023 values, limiting dependence on stage-one forecasts. Prediction artifacts are exported to the repository so every reported number is auditable.